

tf.compat.v1.assert_near Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/assert_near

Assert the condition x and y are close element-wise.

Function Signatures

```
tf.compat.v1.assert_near( x, y, rtol=None, atol=None,
data=None, summarize=None, message=None, name=None )
with tf.control_dependencies([tf.compat.v1.assert_near(x,
y)]): output = tf.reduce_sum(x)
```

Code Examples

```
tf.compat.v1.assert_near(  
    x,  
    y,  
    rtol=None,  
    atol=None,  
    data=None,  
    summarize=None,  
    message=None,  
    name=None  
)
```

```
tf.compat.v1.assert_near(  
    x,  
    y,  
    rtol=None,  
    atol=None,  
    data=None,  
    summarize=None,  
    message=None,  
    name=None  
)
```

```
with tf.control_dependencies([tf.compat.v1.assert_near(x, y)]):  
    output = tf.reduce_sum(x)
```

```
with tf.control_dependencies([tf.compat.v1.assert_near(x, y)]):  
    output = tf.reduce_sum(x)
```

Documentation

Assert the condition x and y are close element-wise.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.debugging.assert_near`

Example of adding a dependency to an operation:

This condition holds if for every pair of (possibly broadcast) elements $x[i]$, $y[i]$, we have

`tf.abs(x[i] - y[i]) <= atol + rtol * tf.abs(y[i]).`

If both x and y are empty, this is trivially satisfied.

The default `atol` and `rtol` is $10 * \text{eps}$, where `eps` is the smallest representable positive number such that $1 + \text{eps} \neq 1$. This is about $1.2\text{e-}6$ in 32bit, $2.22\text{e-}15$ in 64bit, and 0.00977 in 16bit.

See `numpy.finfo`.

Returns

`## numpy compatibility`

Similar to `numpy.testing.assert_allclose`, except tolerance depends on data type. This is due to the fact that TensorFlow is often used with 32bit, 64bit, and even 16bit data.

